WIKIPEDIA

Plant

Plants are mainly multicellular, predominantly photosynthetic eukaryotes of the kingdom **Plantae**. Historically, plants were treated as one of two kingdoms including all living things that were not animals, and all algae and fungi were treated as plants. However, all current definitions of Plantae exclude the fungi and some algae, as well as the prokaryotes (the archaea and bacteria). By one definition, plants form the clade Viridiplantae (Latin name for "green plants"), a group that includes the flowering plants, conifers and other gymnosperms, ferns and their allies, hornworts, liverworts, mosses and the green algae, but excludes the red and brown algae.

Green plants obtain most of their energy from sunlight via photosynthesis by primary chloroplasts that are derived from endosymbiosis with cyanobacteria. Their chloroplasts contain chlorophylls a and b, which gives them their green color. Some plants are parasitic or mycotrophic and have lost the ability to produce normal amounts of chlorophyll or to photosynthesize. Plants are characterized by sexual reproduction and alternation of generations, although asexual reproduction is also common.

There are about 320,000 species of plants, of which the great majority, some 260–290 thousand, produce seeds.[5] Green plants provide a substantial proportion of the world's molecular oxygen,[6] and are the basis of most of Earth's ecosystems. Plants that produce grain, fruit and vegetables also form basic human foods and have been domesticated for millennia. Plants have many cultural and other uses, as ornaments, building materials, writing material and, in great variety, they have been the source of medicines and psychoactive drugs. The scientific study of plants is known as botany, a branch of biology.

Plants

Temporal range: Mesoproterozoic–present

Had'r Archean Proterozoic Pha.

Botany, also called **plant science(s)**, **plant biology** or **phytology**, is the science of plant life and a branch of biology. A **botanist**, **plant scientist** or **phytologist** is a scientist who specialises in this field. The term "botany" comes from the Ancient Greek word βοτάνη (*botanē*) meaning "pasture", "grass", or "fodd...

- **Chlorobionta** Kenrick & Crane 1997
  - Chlorophyta
- **Streptobionta** Kenrick & Crane 1997
  - Klebsormidiophyceae
  - Charophyta (stoneworts)
  - ?Chaetosphaeridiales
  - Coleochaetophyta
  - Zygnematophyta
  - Embryophyta Engler, 1892 (land plants)
    - Marchantiophyta (liverworts)
    - Bryophyta (mosses)
    - Anthocerotophyta (hornworts)
    - †Horneophyta
    - †Aglaophyta

Contents

Definition

All living things were traditionally placed into one of two groups, plants and animals. This classification may date from Aristotle (384 BC – 322 BC), who made the distinction between plants, which generally do not move, and animals, which often are mobile to catch their food. Much later, when Linnaeus (1707–1778) created the basis of the modern system of scientific classification, these two groups became the kingdoms Vegetabilia (later Metaphyta or Plantae) and Animalia (also called Metazoa). Since then, it has become clear that the plant kingdom as originally defined included several unrelated groups, and the fungi and several groups of algae were removed to new kingdoms. However, these organisms are still often considered plants, particularly in popular contexts.